

Chapter 3 : Electrochemistry

PSPM 2011 / 2012

- Q. 1 (a) A galvanic cell undergoes the following reaction:

$$\text{Mg(s)} + \text{Cd}^{2+}(\text{aq}) \longrightarrow \text{Mg}^{2+}(\text{aq}) + \text{Cd(s)}$$
 The magnesium electrode was immersed into a 1.00 M solution of MgSO_4 and the cadmium electrode was immersed into a solution of unknown Cd^{2+} concentration. The potential of the cell was measured to be 1.54 V at 298 K. 9 m
- (i) Write the half cell reaction.
 (ii) Calculate E°_{cell} and determine the spontaneity of the reaction.
 (iii) Calculate the concentration of Cd^{2+} in the cell.
 (iv) Predict what happens to voltage when the Mg^{2+}/Mg half cell is replaced with Zn^{2+}/Zn half cell (1.00 M Zn^{2+})
 $[E^\circ_{\text{Cd}^{2+}/\text{Cd}} = -0.40 \text{ V}; E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}; E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}]$

- Answer (a)
- (i) Anode (oxidation) : $\text{Mg (s)} \longrightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$
 Cathode (Reduction) : $\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Cd (s)}$
- (ii) $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \quad @ \quad E^\circ_{\text{cell}} = E^\circ_{\text{red}} + E^\circ_{\text{ox}}$
 $E^\circ_{\text{cell}} = (-0.40 \text{ V}) - (-2.37 \text{ V})$
 $= +1.97 \text{ V}$
 $\therefore$ Spontaneous reaction
- (iii) $E_{\text{cell}} = E^\circ_{\text{cell}} - (RT / nF) \ln([\text{Mg}^{2+}] / [\text{Cd}^{2+}])$
 $1.54 \text{ V} = (+1.97 \text{ V}) - [(8.314)(298) / (2)(9.65 \times 10^4)] \ln(1 / [\text{Cd}^{2+}])$
 $[\text{Cd}^{2+}] = 2.58 \times 10^{-15} \text{ M}$
 @
 $E_{\text{cell}} = E^\circ_{\text{cell}} - (0.0592 / n) \log([\text{Mg}^{2+}] / [\text{Cd}^{2+}])$
 $1.54 = 1.97 - (0.0592 / 2) \log(1 / [\text{Cd}^{2+}])$
 $[\text{Cd}^{2+}] = 2.97 \times 10^{-15} \text{ M}$
 (* final answer $2.80 \times 10^{-15} \text{ M}$ $3.0 \times 10^{-15} \text{ M}$)
- (iv) • The voltage will be reduced / decreased
 • Because E°_{cell} for the reaction is smaller than 1.97 V
 @
 • $E^\circ_{\text{Zn}^{2+}/\text{Zn}}$ closer to $E^\circ_{\text{Mg}^{2+}/\text{Mg}}$ in standard reduction potential table.

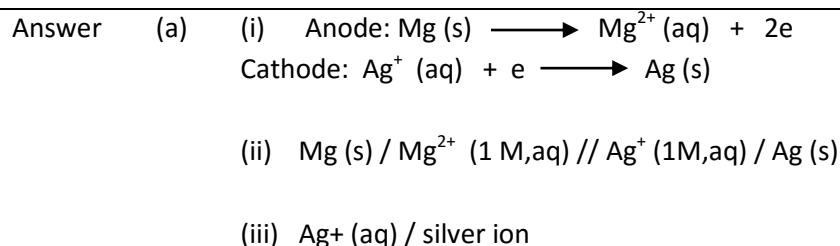
PSPM 2012 / 2013

- Q. 2 (a) A galvanic cell consists of a magnesium electrode immersed in 1.0 M magnesium nitrate solution and a silver electrode immersed in 1.0 M silver nitrate solution and a salt-bridge.

$$[E^{\circ} \text{Mg}^{2+}/\text{Mg} = -2.36 \text{ V}, E^{\circ} \text{Ag}^{+}/\text{Ag} = +0.80 \text{ V}]$$

4 m

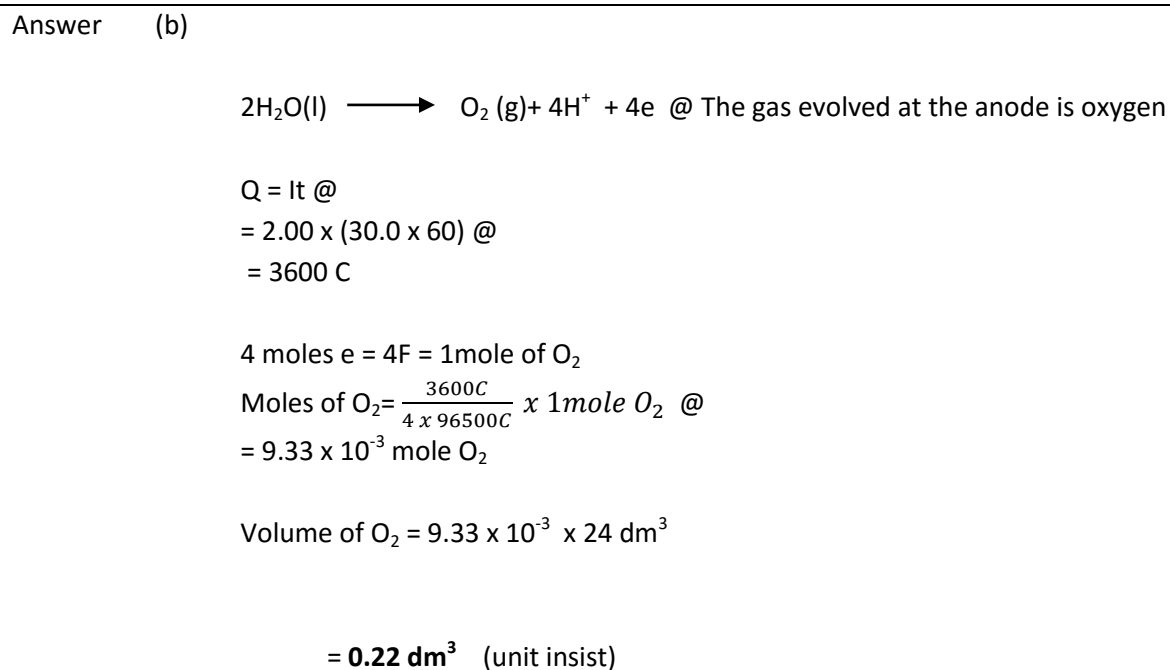
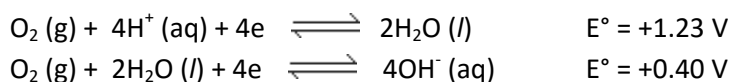
- (i) Write the half reaction at anode and cathode.
 (ii) Write the cell notation for the above reaction.
 (iii) State the oxidizing agent.



- Q. 2 (b) A current of 2.00 A is passed for 30 minutes through a galvanic cell consisting of graphite electrodes and a solution of potassium sulphate at 25 °C. Calculate the volume of oxygen gas evolved at the anode at 1 atm.

5 m

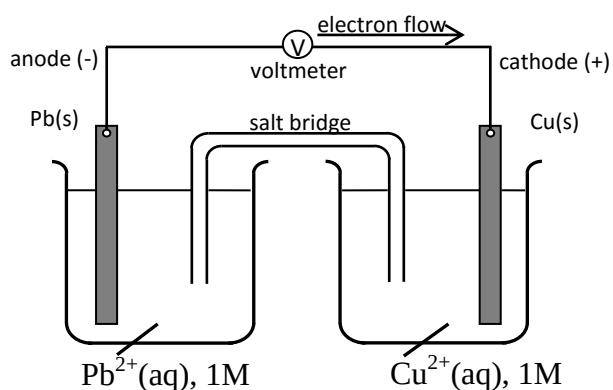
Given:



PSPM 2013 / 2014

- Q. 3 (a) Given $E^\circ \text{Pb}^{2+} | \text{Pb} = -0.13 \text{ V}$, $E^\circ \text{Cu}^{2+} | \text{Cu} = +0.34 \text{ V}$ for galvanic cell at standard conditions.
- (i) Sketch the cell, indicating the flow of electrons, anode and cathode terminals.
- (ii) Write the cell notation
- (iii) Calculate E_{cell} for the reaction if the concentration of Pb^{2+} and Cu^{2+} is 1.0 M and 0.1 M respectively.

Answer (a) (i)



Anode + cathode – 1M

Electrolyte anode & cathode concentration – 1M

- (ii) Salt bridge + voltmeter – 1M
Electron flow – 1M



$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} @ +0.34 \text{ V} - (-0.13 \text{ V}) \\ = +0.47 \text{ V}$$

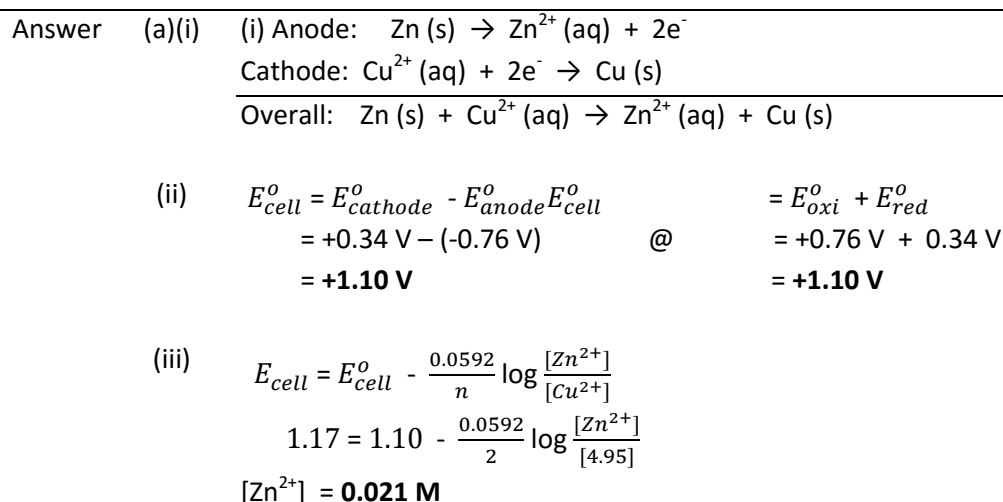
Using Nernst Equation;

$$E_{\text{cell}} = E^\circ - \frac{RT}{nF} \ln \left(\frac{\text{Pb}^{2+}}{\text{Cu}^{2+}} \right) @ \\ + 0.47 \text{ V} - \frac{(8.314 \text{ J mol}^{-1} \text{ K}^{-1})(298.15 \text{ K})}{(2)(96500 \text{ J V}^{-1} \text{ mol}^{-1})} \ln \left(\frac{1.0 \text{ M}}{0.1 \text{ M}} \right) \\ = +0.44 \text{ V}$$

PSPM 2014 / 2015

- Q. 4 (a) A voltaic cell is set up using zinc and copper half-cells.

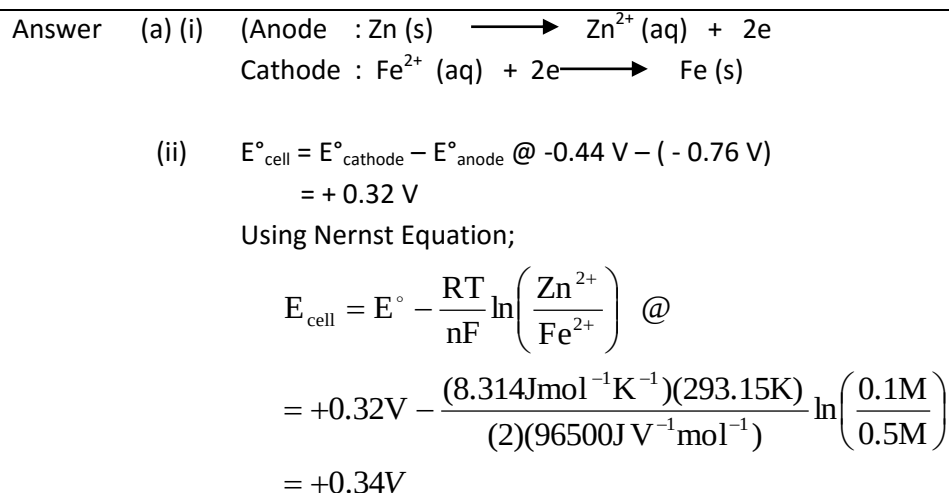
$$\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}) \parallel \text{Cu}^{2+}(\text{aq}) \mid \text{Cu(s)}$$
- (i) Write the half reaction equations at both electrodes and the overall cell reaction equation. 8 m
- (ii) Calculate the cell potential under standard condition
 If the cell potential is 1.17 V and the concentration of Cu^{2+} ion is 4.95 M,
- (iii) Calculate the concentration of Zn^{2+} ion.
 [Given: $E_{\text{Cu}^{2+}|\text{Cu}}^{\circ} = +0.34 \text{ V}$, $E_{\text{Zn}^{2+}|\text{Zn}}^{\circ} = -0.76 \text{ V}$]



PSPM 2015 / 2016

- Q. 5 (a) Consider the cell notation below for a galvanic cell at standard conditions.

$$\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}, 1.0 \text{ M}) \parallel \text{Fe}^{2+}(\text{aq}, 0.5 \text{ M}) \mid \text{Fe(s)}$$
 7 m
- (i) Write the half reaction that occurs at both electrodes.
- (ii) Calculate E_{cell} for the electrochemical system.
 [Given: $E_{\text{Fe}^{2+}|\text{Fe}}^{\circ} = -0.44 \text{ V}$ and $E_{\text{Zn}^{2+}|\text{Zn}}^{\circ} = -0.76 \text{ V}$]



PSPM 2016 / 2017

- Q. 6 (a) A galvanic cell is constructed by connecting a nickel plate that was immersed into 1.50M $\text{Ni}(\text{NO}_3)_2$ solution and an aluminium plate that was immersed into a solution containing x M of Al^{3+} ions. If the cell voltage is 1.40V, 5 m

- (i) Calculate the standard cell potential.
(ii) Determine the value of x.

[Given: $E^\circ \text{Ni}^{2+} | \text{Ni} = -0.25\text{V}$, $E^\circ \text{Al}^{3+} | \text{Al} = -1.66\text{V}$]

Answer (a) (i) Anode : $2\text{Al}(\text{s}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 6\text{e}^-$

Cathode : $3\text{Ni}^{2+}(\text{aq}) + 6\text{e}^- \rightarrow 3\text{Ni}(\text{s})$

Overall : $2\text{Al}(\text{s}) + 3\text{Ni}^{2+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{Ni}(\text{s})$

$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \\ &= -0.25 - (-1.66) \\ &= \mathbf{+1.41\text{ V}} \end{aligned}$$

(ii)

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0592}{n} \log \frac{[\text{Al}^{3+}]^2}{[\text{Ni}^{2+}]^3}$$

$$1.40 = 1.41 - \frac{0.0592}{6} \log \frac{[x]^2}{[1.50]^3}$$

$$x = \mathbf{5.90\text{ M}}$$

- Q. 6 (b) Electroplating is one of the techniques used as a protective coating for metals. A student is requested to electroplate the iron nail with tin. If a current is passed through the electroplating solution, write the half-cell equations at the anode and cathode. Predict what will occur at both terminals. 4 m

Answer (c) Cathode : $\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$

Anode : $\text{Sn}(\text{s}) \rightarrow \text{Sn}^{2+}(\text{aq}) + 2\text{e}^-$

Cathode : Mass of cathode/nail increase @ Tin deposited on the nail @ nail becomes thicker

Anode : Mass of anode reduced @ Mass of tin strip reduced @ tin becomes thinner

PSPM 2017 / 2018

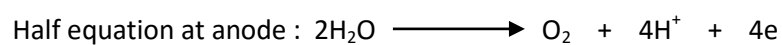
- Q. 7 (a) An aqueous solution of lead(II) nitrate was electrolysed using a current of 2.0 A for 30 minutes. Determine the volume of gas evolved at the anode at room temperature . 6 m

Answer (a)

$$Q = It$$

$$= 2.0 \times (30 \times 60)$$

$$= 3600 \text{ C}$$



$$4F \equiv 1 \text{ mol O}_2$$

$$4(96500 \text{ C}) \equiv 1 \text{ mol O}_2$$

$$3600 \text{ C} \equiv 3600 / 4(96500)$$

$$\equiv 9.326 \times 10^{-3} \text{ mol O}_2$$

$$\begin{aligned} \text{Volume of oxygen at room temperature} &= 9.326 \times 10^{-3} \times 24 \text{ dm}^3 \text{ mol}^{-1} \\ &= \mathbf{0.2238 \text{ dm}^3} \end{aligned}$$

PSPM 2018 / 2019

- Q. 8 (a) A galvanic cell is constructed based on the notation below:

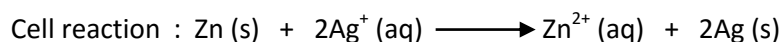
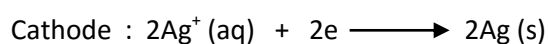
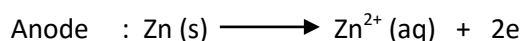


The cell potential is 1.58 V when concentration of Zn^{2+} and Ag^+ are x M and 0.15 M, respectively. Calculate the value of x .

4m

[Given: $E^\circ_{\text{cell}} = +1.56 \text{ V}$]

Answer



$$n = 2$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0592}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$$

$$1.58 = +1.56 - \frac{0.0592}{2} \log \frac{(x)}{(0.15)^2}$$

$$[x] = [\text{Zn}^{2+}] = 4.75 \times 10^{-3} \text{ M}$$

- (b) A charge of 0.025 F was passed through an aqueous solution of M^{n+} using a current of 2.00 A. An amount of 0.79 g of **M** was deposited at the cathode.

6m

[Given molar mass of **M** is 63.2 g mol^{-1}]

- (i) Calculate the time taken in minutes for the deposition of **M**.

Answer

$$Q = 0.025 \times 96500$$

$$= 2412.5 \text{ C}$$

$$Q = I \times t$$

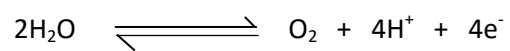
$$t = 2412.5 / 2.00$$

$$= 1206.25 \text{ s}$$

$$= 20 \text{ minutes}$$

- (ii) Calculate the volume of oxygen gas liberated at the anode at STP.

Answer



From equation, $4 \text{ mol e}^- \equiv 1 \text{ mol O}_2$

$$0.025 \text{ mol e}^- \equiv 0.025 / 4 \times 1 = 6.25 \times 10^{-3} \text{ mol O}_2$$

$$\begin{aligned} \text{Volume of O}_2 &= 6.25 \times 10^{-3} \times 22.4 \\ &= \mathbf{0.14 \text{ dm}^3} \end{aligned}$$